Supplementary material for: A physics-informed digital twin for tube-life-aware operation of industrial steam methane reformers under uncertainty

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S1. Model implementation detail

Effectiveness factors. Intraparticle diffusion limitation is severe in reformer catalysts. Rather than solving the pellet problem, the constant effectiveness factors adopted by Latham et al. [3] following Wesenberg and Svendsen [4] are used: $\eta_j = 0.1$ along the tube and 0.05 in the first tube section (10 % of the length here), where the feed contains little hydrogen and the rate expressions are stiff.

Higher alkanes at the inlet. Higher alkanes in the feed (C_2–C_5) are converted at the inlet by the overall rule used by Latham [3],

$$\mathrm{C_kH_{2k+2}} + \tfrac{k-1}{3}\,\mathrm{H_2O} \rightarrow \tfrac{k-1}{3}\,\mathrm{CO} + \tfrac{2k+1}{3}\,\mathrm{CH_4}, \tag{1}$$

whose enthalpy is small because the endothermic reforming and the exothermic hydrogenolysis nearly cancel.

Equilibrium constants and the hydrogen floor. The equilibrium constants computed from Gibbs energies of formation agree with the common empirical expressions within 7 % between 800 and 1100 K. A floor of 10^{-6} bar on $p_{\mathrm{H_2}}$ avoids the singularity of the rate expressions at a hydrogen-free inlet.

Heat-release profile and furnace-side convection. The heat-release density $r(z)$ of the long-furnace model is a downward-opening parabola that vanishes at $z = L_q L$, releases the fraction α_{top} of the heat in the top fifteenth of the furnace height and integrates to $(1 - f_{\mathrm{loss}})\,Q_{\mathrm{comb}}$, with $f_{\mathrm{loss}} = 0.02$ for casing losses; this is the continuous form of the sectional heat-release profile of Latham [3]. The furnace-side convective coefficient h_c is taken from the Dittus–Boelter correlation on the flue-gas flow area, scaled by the factor $f_{\mathrm{ctube}} = 0.38$ fitted by Latham et al. [3]; at furnace-gas temperatures the convective term is a minor correction to the radiative flux. The combustion heat Q_{comb} and the flue-gas flow and composition follow from the lower heating value and complete combustion of the fuel gas and pressure-swing-adsorption off-gas with the specified excess air.

Coupled solution. The outer-wall temperature at each height is found by a bracketed root search on the flux balance, evaluated inside the right-hand side of the ODE system, so that the furnace and tube equations are integrated together in a single downward sweep.

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Thermal stress across the tube wall. The elastic thermal stress from the radial temperature gradient, $E\alpha\Delta T_{\mathrm{wall}}/[2(1-\nu)]$ with $E = 150$ GPa, $\alpha = 18 \times 10^{-6}$ K^{-1} at 900 °C and $\nu = 0.3$, is computed from the wall drop of the calibrated tube rather than estimated. Averaged over the heated length the drop is 32 K and the stress 63 MPa. At the hot spot, 0.43 of the heated length, they are 57 K and 110 MPa; the gradient is steepest slightly higher up the tube, at 0.35, where they reach 63 K and 122 MPa. The length-average is the figure usually quoted for reformer tubes, and the hot spot is roughly twice it. Even the length-average stress is about five times the 12.9 MPa hoop stress, and at the hot spot the ratio approaches nine.

Thermal time constants. The thermal time constant of the tube wall, $\rho c\, t^2/\lambda$, is of the order of one minute; that of the refractory lining is of the order of a day.

Calibration procedure. The nine residuals per plant case are the outlet temperature, the outlet pressure, the outlet wet composition, the flue-gas outlet temperature and the two wall temperatures. The trust-region-reflective algorithm with bounds is started from a Latin-hypercube set of initial points to guard against local minima, and the parameter covariance is obtained from the Jacobian at the optimum with t-based 95 % intervals.

S2. Operating space and campaign design

Table S1: Operating space sampled in the design, sensitivity, surrogate and optimization campaigns. The base column is the calibrated Plant A state. Ranges are motivated by industrial practice and by the operating points of the sources used for verification and calibration.

Variable	Unit	Min	Max	Base	Basis
steam-to-carbon ratio	–	2.5	4.0	2.89	carbon-formation limit on Ni catalyst below about 2.5; upper value typical of hydrogen plants; 2.7–3.0 in the reference plant [3], 3.36 in [2]
feed per tube	fraction of base	0.60	1.10	1.00	industrial turndown of about 50–60 %; 110 % short-term overload
specific firing Q_{comb}/feed	fraction of base	0.85	1.15	1.00	burner control range around the plant value
inlet temperature	K	800	900	884.5	884.5 K in the reference plant, 793 K in [2]; cracking in the preheat coil above about 920 K
inlet pressure	bar	25	33	30.1	industrial range 20–40 bar; 28–30 bar in the reference plant, 29 bar in [2]
catalyst activity	–	0.10	0.30	0.20	0.20 fitted to the reference plant for a mid-campaign catalyst [3]; fresh higher, end-of-campaign lower
excess air	%	5	25	21.6	industrial reformer furnaces 5–20 %; upper value extended so that the calibrated base lies inside the sampled space

The campaigns evaluated on this space are: an open-loop Latin-hypercube design of 2000 points (surrogate training); a Saltelli design of 4096 points for the Sobol indices and as an independent surrogate test set; a closed-loop 15×15 grid in steam-to-carbon ratio and load at constant outlet temperature (Fig. 5 of the main text); twenty scenario-years of 8760 hourly states; an NSGA-II run of 200 individuals over 300 generations; and a scrambled Sobol Monte Carlo design of 2048 samples propagated through the physics model at eight operating points, with 1024 further samples per uncertainty group for the variance decomposition. All designs use seed 0. The complete pipeline runs in 35 minutes on a ten-core laptop.

Basis of the hot-spot temperature limit. The optimization constrains the hot-spot temperature to 1193 K (920 °C), close to the 925 °C design tube-metal temperature used by Yeh [7] for an HP-Nb tube and 46 K above the calibrated base hot spot.

Enthalpy basis of the steam credit. The steam duty charged at $\alpha < 1$ is the heat needed to raise the process steam from liquid water at 25 °C to vapor at the inlet temperature, plus the heat needed to preheat the dry feed to the inlet temperature.

Table S2: Definitions of the six annual scenarios of Section 2.7 of the main text. Each is an 8760-hour hourly history evaluated under one of the two control modes.

Scenario	Control mode	Definition
S1 steady base	either	constant operation at the calibrated Plant A state
S2 daily load-following	either	full load 06–22 h, 70 % at night, ramps of 0.10 per hour
S3 renewable-following	either	bounded Ornstein–Uhlenbeck load process between 0.60 and 1.10 with mean 0.85
S4 over-firing events	either	24 events per year of $+10\,\%$ firing for 2 h; six events of $+25\,\%$ for 1 h; a flame-impingement proxy adding 40 K to the hot spot for 24 h per year
S5 catalyst aging	both, separately	activity falling linearly from 0.30 to 0.10 over four years
S6 steam-to-carbon	either	steady operation at steam-to-carbon 2.5 and at 3.5

Table S3: Uncertainty groups propagated in RQ4 and the distribution assigned to each input.

Group	Inputs	Distribution
kinetics	k_1–k_3, K_{CO}, $K_{\mathrm{H_2}}$, $K_{\mathrm{CH_4}}$, $K_{\mathrm{H_2O}}$	normal, from the 95 % intervals reported by Xu and Froment [1]
	effectiveness factors	log-uniform factor 0.5–2
	catalyst activity	uniform ± 0.03
heat transfer	F_{gt}, L_q, f_{htg}	multivariate normal with the joint covariance of the calibration
creep	scatter on $\log_{10} t_r$	normal, 0.181 decades, from the data-sheet band
	Larson–Miller constant C	uniform within ± 1 of 18.6
	wall thickness	uniform 12–18 mm
	tube conductivity	uniform $\pm 15\,\%$
measurement	hot-spot temperature offset	uniform ± 10 K on the pyrometer reference

The flame-length variant is a bound. The structural variant $L_q\,\mathrm{load}^{0.5}$ of the uncertainty analysis is a bound, not an expectation: the length of a turbulent jet diffusion flame is nearly independent of the jet velocity for a given burner [5], so that reducing the fuel flow through all burners leaves the release profile essentially unchanged (the nominal case), whereas taking burner rows out of service at turndown, a common practice, redistributes the heat release in a way that the published data cannot constrain.

S3. Surrogate performance

The eight surrogate targets are the hot-spot temperature $T_{\mathrm{wo,max}}$ and its axial position, the outlet temperature and pressure, the dry methane slip, the net hydrogen per tube, the absorbed duty and $\log_{10} t_r$ at the hot spot. Three model families are compared with scikit-learn [6]: Gaussian-process regression with a Matérn 5/2 kernel on a 1000-point subset, histogram gradient boosting and a multilayer perceptron. Each family is fitted twice, once on the seven operating inputs alone and once with two physics-derived features added: the equilibrium methane conversion at outlet conditions and the specific duty.

Table S4: Best surrogate per target ($N_{\mathrm{train}} = 2000$ Latin-hypercube points, $N_{\mathrm{test}} = 4096$ Saltelli points): model family and feature set, test R^2 and RMSE, five-fold cross-validated RMSE, and empirical coverage (PICP) and mean width (MPIW) of the nominal 90 % CV+ conformal interval. "physics" denotes the feature set augmented with the equilibrium methane conversion and the specific duty.

Target	Unit	Model	R^2	RMSE	CV RMSE	PICP_{90} / MPIW_{90}
$T_{\mathrm{wo,max}}$	K	GP, physics	1.000	0.113	0.133	0.957 / 0.25
z/L at $T_{\mathrm{wo,max}}$	–	MLP, base	0.644	0.0319	0.0522	0.971 / 0.050
T_{out}	K	GP, physics	1.000	0.0182	0.0254	0.940 / 0.061
p_{out}	bar	GP, base	1.000	2.87×10^{-4}	5.63×10^{-4}	0.961 / 1.0×10^{-3}
CH_4 slip (dry)	%	GP, physics	1.000	1.25×10^{-3}	1.42×10^{-3}	0.953 / 4.0×10^{-3}
H_2 per tube	$\mathrm{kmol\,h^{-1}}$	GP, base	1.000	8.47×10^{-4}	1.16×10^{-3}	0.946 / 2.9×10^{-3}
absorbed duty	W	GP, base	1.000	3.9×10^{3}	4.14×10^{3}	0.949 / 1.2×10^{4}
$\log_{10} t_r$ at hot spot	–	GP, base	0.9999	4.94×10^{-3}	5.16×10^{-3}	0.958 / 9.56×10^{-3}

The Gaussian process dominates the other families by one to two orders of magnitude in RMSE on every continuous target, which is expected because the physics model is smooth and deterministic and the surrogate is effectively

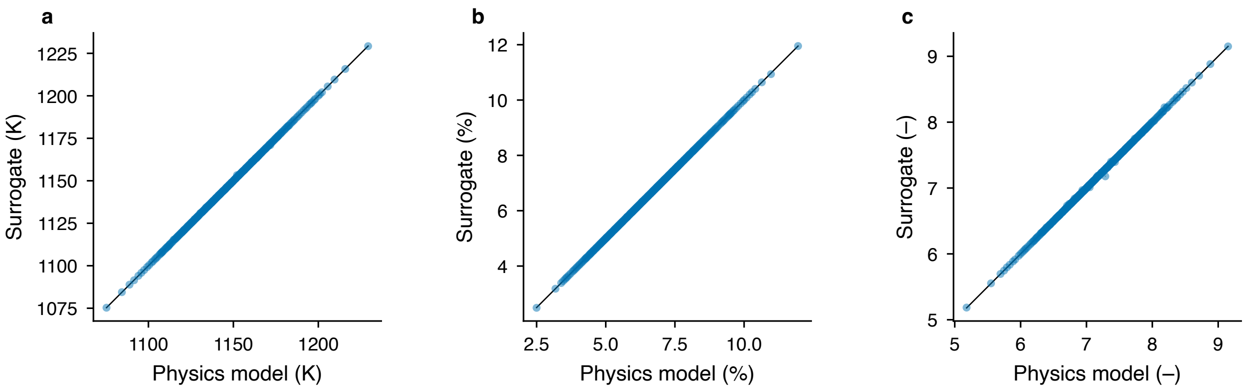

Figure S1: Parity of the selected surrogates against the physics model on the independent Saltelli test set for (a) the hot-spot temperature, (b) dry methane slip and (c) $\log_{10} t_r$ at the hot spot, with nominal 90 % CV+ conformal intervals.

interpolating it; histogram gradient boosting is the weakest, with an RMSE of about 3 K on the hot-spot temperature. The location of the hot spot is the only target that is not learnable: it is bimodal, sitting near $z/L = 0.43$ for most inputs and moving to the outlet plateau for others, and all three families give R^2 below 0.65. It is reported for completeness and is not used in the life calculation, which requires only the maximum. Prediction takes 38 μs per point against 0.25 s for the physics model.

S4. Digitisation of the verification curves

Figure 3 of Xu and Froment [2] reports six axial profiles for one industrial reformer tube on three vertical axes: methane and CO_2 conversion on the left axis, total pressure in bar and the process-gas, inner-wall and outer-wall temperatures in kelvin on two right axes. The figure was digitized with WebPlotDigitizer (version 5) from a 600 dpi scan of the printed page. Three independent axis calibrations were defined on the same image, one per vertical axis, each anchored on two gridline labels of the horizontal axis (0 and 10 m) and two labels of the corresponding vertical axis (0 and 0.8 for conversion, 24 and 29 bar for pressure, 800 and 1150 K for temperature). Points were placed manually along each curve at intervals of roughly 0.05 m of the tube length, with additional points concentrated at the discontinuity at the end of the heated length ($z = 11.12$ m), giving 118–261 points per curve.

The left axis of the original figure is labelled $x/(1+\zeta)$, where $\zeta = 0.056$ is the CO_2/CH_4 molar ratio of the feed; the digitized ordinates of the two conversion curves are therefore multiplied by $1+\zeta$ to obtain conversions referred to the equivalent methane feed. The residual uncertainty of the digitisation, estimated from the spread of repeated point placements on the gridlines, is about 2 K in temperature, 0.03 bar in pressure and 0.005 in conversion, which is small compared with the model residuals reported in the main text. The digitized curves, the axis calibrations and a tidy merged file with provenance metadata are in `data/literature_validation/digitized/` of the repository.

Two features of the digitized data are recorded but not corrected. The digitized process-gas temperature begins at 764 K whereas Table 2 of the same paper states an inlet temperature of 793 K, so all temperature comparisons in the main text exclude $z < 0.3$ m. Both conversion curves begin at about 0.057 rather than at zero; this value lies between the inlet conversion implied by the higher-alkane rule used here (0.018) and the value obtained if all C_2–C_5 carbon is converted to CO at the inlet (0.091), and the conversion increment $x(z) - x(0.5\ \mathrm{m})$ is therefore used as the primary comparison metric. The offset is consistent with a steep initial rise that the printed figure does not resolve, and it accounts for the residual of about 0.04 in absolute conversion reported in the main text.

S5. Verification and calibration detail

The pressure RMSE of the verification runs rises from 0.04 bar with the Leva–Grummer configuration to 0.25 bar with the matched film coefficient. This is not a deterioration of the friction model, which is unchanged between

Table S5: Verification of the tube model against the digitized Fig. 3 of Xu and Froment [2]: RMSE of methane conversion $x_{\mathrm{CH_4}}$, of the conversion increment $x(z) - x(0.5\ \mathrm{m})$, of the process-gas and inner-wall temperatures ($z \geq 0.3$ m) and of pressure. Bed voidage 0.526 and equivalent particle diameter 9.44 mm fitted to the pressure profile.

Configuration	RMSE $x_{\mathrm{CH_4}}$	RMSE Δx	RMSE T_g (K)	RMSE T_{wi} (K)	RMSE p (bar)
Leva–Grummer, $f_{\mathrm{htg}} = 1$, $\eta = 0.1$	0.160	–	75.4	37.5	0.04
matched $\alpha_i = 348\ \mathrm{W\,m^{-2}\,K^{-1}}$, $\eta = 0.1$	0.037	0.020	13.7	16.7	0.25
matched α_i, $\eta = 0.05/0.1$ (adopted)	0.039	0.015	14.0	16.7	0.25
matched $\alpha_i(z)$ profile, $\eta = 0.05/0.1$	0.046	0.017	17.6	16.9	0.26

Back-calculated α_i: median 348, interquartile range 313–363 $\mathrm{W\,m^{-2}\,K^{-1}}$ over 0.5–11 m.

Table S6: Calibrated furnace and heat-transfer parameters on Plants A, B and C2 with 95 % t-intervals from the Jacobian covariance. The four-parameter fit has α_{top} at its bound with $r(L_q, \alpha_{\mathrm{top}}) = 0.985$; the adopted fit fixes $\alpha_{\mathrm{top}} = 0.182$.

Fit	Parameter	Value	95 % interval	s.e.
four parameters	F_{gt}	0.316	[0.286, 0.346]	0.014
	L_q	0.497	[0.329, 0.665]	0.081
	α_{top}	0.400	[−0.261, 1.061]	0.32
	f_{htg}	1.74	[1.427, 2.051]	0.15
	χ^2 / dof	30.5 / 23	reduced 1.32	
α_{top} fixed (adopted)	F_{gt}	0.323	[0.309, 0.338]	0.0072
	L_q	0.456	[0.434, 0.478]	0.011
	f_{htg}	1.83	[1.617, 2.052]	0.11
	χ^2 / dof	34.9 / 24	reduced 1.46	

the two: the bed voidage and equivalent particle diameter were fitted to the pressure profile under the hotter Leva–Grummer gas, and the cooler gas of the matched run has a higher density and a lower velocity at the same mass flow.

S6. Steam-credit sweep

Table S7: Steam-credit sweep of the iso-production (H_2 = base ±1 %) life–fuel front. $\alpha = 1$ counts firebox fuel only; $\alpha = 0$ charges steam raising and feed preheat in full. All rows are physics-verified.

α	Regime	S/C	Load	Firing	Excess air (%)	Rel. life cost	Total fuel vs base (%)	$T_{\mathrm{wo,max}}$ (K)
0.00	min life	4.00	0.96	0.930	24.6	0.094	+4.1	1103
0.00	knee	3.36	0.97	0.851	5.0	0.783	−6.3	1142
0.00	min fuel	2.64	1.00	0.887	5.0	3.647	−8.7	1172
0.25	min life	4.00	0.96	0.908	24.7	0.093	+0.5	1103
0.25	knee	3.84	0.94	0.850	5.0	0.416	−7.0	1130
0.25	min fuel	2.96	0.99	0.866	5.0	1.706	−8.9	1157
0.50	min life	4.00	0.96	0.908	24.8	0.093	−2.8	1103
0.50	knee	4.00	0.95	0.850	11.0	0.195	−8.4	1116
0.50	min fuel	3.69	0.94	0.850	5.0	0.504	−10.4	1134
0.75	min life	4.00	0.96	0.909	24.9	0.092	−6.8	1103
0.75	knee	4.00	0.96	0.850	14.3	0.143	−12.0	1111
0.75	min fuel	4.00	0.93	0.850	5.0	0.342	−14.6	1127
1.00	min life	4.00	0.96	0.902	23.7	0.096	−12.7	1104
1.00	knee	4.00	0.96	0.850	14.3	0.144	−18.0	1111
1.00	min fuel	4.00	0.93	0.850	5.0	0.341	−20.4	1127

S7. Sensitivity indices

Beyond the three leading drivers quoted in the main text, inlet temperature, catalyst activity and load each contribute below 0.1 to the variance of the hot-spot temperature, and inlet pressure is negligible for it. Excess air matters

Table S8: Sobol first-order (S_1) and total-effect (S_T) indices with 95 % confidence half-widths from a Saltelli design ($N = 256$) over the operating space of Table S1.

Input	S_1 ($T_{\mathrm{wo,max}}$)	S_T ($T_{\mathrm{wo,max}}$)	S_1 ($\log_{10} t_r$)	S_T ($\log_{10} t_r$)	S_1 (CH_4 slip)	S_T (CH_4 slip)
steam-to-carbon	0.29 ± 0.09	0.28 ± 0.05	0.25 ± 0.09	0.25 ± 0.05	0.04 ± 0.03	0.04 ± 0.01
load	0.03 ± 0.03	0.03 ± 0.01	0.03 ± 0.03	0.03 ± 0.01	0.19 ± 0.07	0.19 ± 0.03
firing	0.34 ± 0.09	0.35 ± 0.06	0.28 ± 0.08	0.29 ± 0.06	0.54 ± 0.12	0.54 ± 0.10
inlet T	0.06 ± 0.05	0.06 ± 0.01	0.05 ± 0.04	0.05 ± 0.01	0.08 ± 0.06	0.09 ± 0.02
inlet p	0.02 ± 0.02	0.02 ± 0.00	0.17 ± 0.07	0.17 ± 0.03	0.03 ± 0.03	0.03 ± 0.01
activity	0.07 ± 0.04	0.08 ± 0.02	0.06 ± 0.04	0.06 ± 0.01	0.00 ± 0.01	0.00 ± 0.00
excess air	0.19 ± 0.06	0.19 ± 0.03	0.16 ± 0.06	0.16 ± 0.03	0.11 ± 0.04	0.11 ± 0.02

because at fixed fuel it lowers the adiabatic flame temperature and raises the flue-gas flow, both of which move heat release down the tube. First-order and total-effect indices agree within their confidence intervals for every input, so interactions are weak over the operating space and the influence of each variable can be read from one-at-a-time maps. The methane slip is governed by the firing factor ($S_T = 0.54$) and the load, as expected for an approach-to-equilibrium quantity.

S8. Variance decomposition

Table S9: Variance decomposition at the S1 base point: one-group-at-a-time share of the joint variance ($N = 1024$ per group) and group total-effect Sobol index computed on a Gaussian-process surrogate of the joint sample ($N = 2048$).

Group	share, $\log_{10} t_r$	S_T, $\log_{10} t_r$	share, $T_{\mathrm{wo,max}}$	S_T, $T_{\mathrm{wo,max}}$
kinetics	0.085	0.157	0.513	0.620
heat transfer	0.018	0.080	0.111	0.302
creep	0.859	0.779	0.211	0.093
measurement	0.034	0.025	0.202	0.092
sum	0.997		1.037	

The one-group-at-a-time shares sum to unity within 0.04, so the four groups act almost additively and the interaction between them is negligible at the base point. The share of the creep group in the hot-spot temperature (0.211) is not a feedback of the creep model on the thermal solution, which does not exist, but the effect of the wall thickness, which belongs to the creep group because it also enters the hoop stress and which changes the conduction drop across the tube wall.

S9. Supplementary figures

S10. Extension to the tube population

The life results of the main text are statements about one average tube. The reference furnace has 7 rows of 48 tubes, and the standard deviation of the wall temperature across that population at a given elevation is 15–22 K [3]. Because the damage rate is convex in wall temperature, the mean damage rate of the population is larger than the damage rate evaluated at the mean temperature, and the tube that decides when the furnace is retubed is not the average tube but the hottest one.

The extension assigns each of the $N_t = 336$ tubes a wall-temperature offset $\Delta T_i \sim \mathcal{N}(0, \sigma^2)$, held fixed over the year rather than resampled hourly, and evaluates the rupture time at $T_{\mathrm{wo,max}} + \Delta T_i$ on the same master curve and at the same hoop stress. Firing, feed and pressure are common to the bundle, so the hoop stress is common as well and the process model is not re-entered per tube. Table S10 reports the result at the calibrated base as ratios to the average tube; absolute lives are not quoted, for the reasons given in Section 4.3 of the main text.

The sampled population mean sits within one Monte Carlo standard error of the closed-form result at every σ. Repeating the sampling with 4×10^5 tubes isolates the residual bias of the closed form at 1.5–4.5 %, systematically

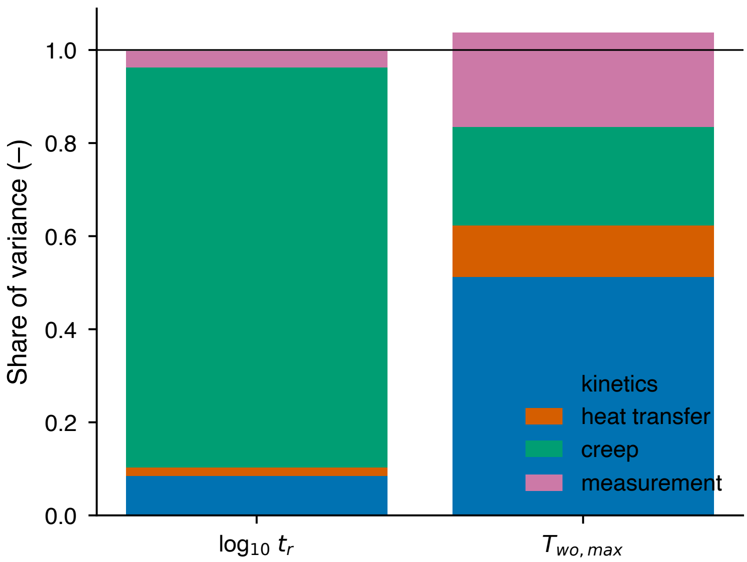

Figure S2: One-group-at-a-time shares of the variance of $\log_{10} t_r$ and of the hot-spot temperature at the S1 base point; numerical values in Table S9.

Table S10: Damage-rate multipliers relative to the average tube for a population of 336 tubes at the calibrated base (S1, firing solved to hold the measured outlet temperature), as a function of the population standard deviation σ. The analytic column is the closed-form lognormal result $\exp\left[(a \ln 10\, \sigma)^2/2\right]$ with $a = 0.0225$ decades per kelvin, the local slope of $\log_{10} t_r$ against wall temperature at this point. The hottest-tube column is the maximum of one bundle; the last two columns give its mean, standard deviation and 5–95 % range over 200 independent bundles, since a single furnace is one draw.

σ (K)	population mean	analytic	hottest decile	p95 tube	hottest tube	over 200 bundles	5–95 %
15.0	1.29	1.35	3.86	3.32	9.9	9.0 ± 2.6	5.6–14.1
18.5	1.49	1.58	5.34	4.37	16.5	14.8 ± 5.2	8.3–25.3
22.0	1.75	1.92	7.43	5.73	27.2	24.2 ± 10.2	12.2–44.7

negative because the slope a falls as $1/T^2$, so the hot tail gains slightly less life consumption than a constant-slope lognormal assumes. The closed form is therefore a usable approximation for the population mean, and the reported values are the sampled ones.

Two qualifications bound the hottest-tube figures. The scatter is assumed normal, and the hottest tube is the extreme of that assumed tail, so its multiplier inherits whatever the tail shape is; the hottest decile is the more robust statistic and is quoted in the main text alongside it. And the maximum of a bundle is itself a random variable, which is why the spread over 200 bundles is given: two nominally identical furnaces differ by a factor of three in the life of their worst tube at $\sigma = 18.5$ K. The measurement group of the uncertainty analysis of the main text (± 10 K) is a different quantity, an epistemic offset of the pyrometer reference common to all tubes rather than the spread between them, and the two are not combined. A per-tube wall-temperature distribution measured by a pyrometer survey would replace the assumed normal and would also bound the tail directly.

The population correction is close to common across operating regimes: over the Pareto regimes of Table 3 of the main text it ranges from 1.42 for the hottest regime (maximum hydrogen) to 1.61 for the coolest (minimum life overall), so it compresses the spread between regimes slightly without reordering them. Under the population mean the ranking of the RQ3 regimes by life consumption per kmol H_2 is identical to the ranking of the average tube, and it remains identical under the hottest decile. Among the twenty RQ2 scenario rows, two change rank under the population mean and three under the hottest decile; all of them lie between ranks 13 and 15 and the pairs that exchange places are S3 against S4c and S2 against S4c, separated by 0.47 % and 0.78 % on the average tube. Every conclusion drawn in the main text — the ageing ratio, the steam-to-carbon comparison and the knee against the base — keeps both its order and its margin.

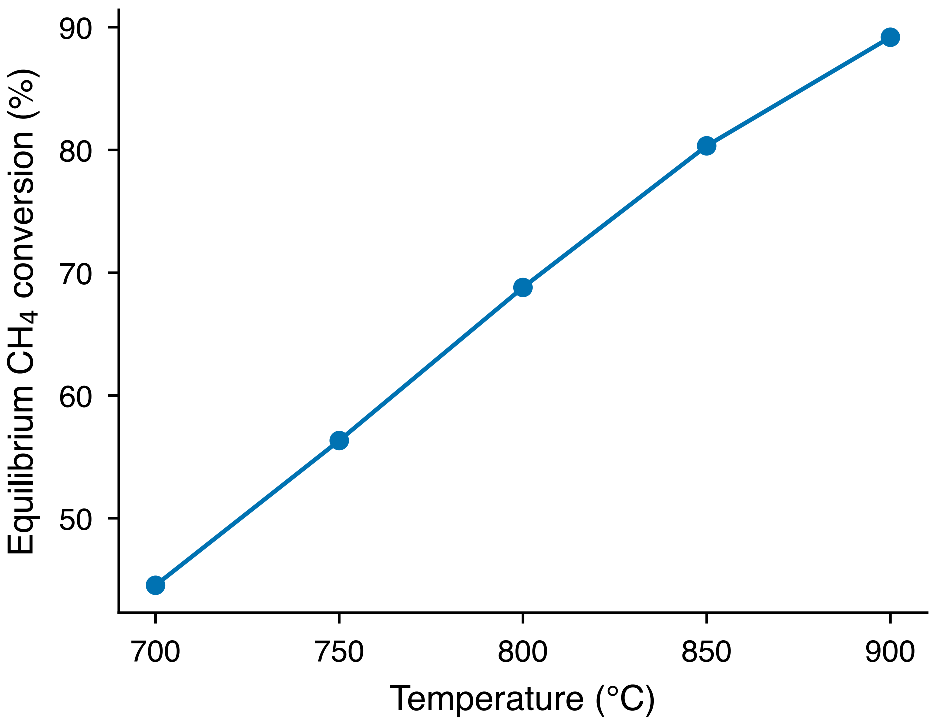

Figure S3: Equilibrium methane conversion of a CH_4/H_2O mixture at 25 bar and a steam-to-carbon ratio of 3, computed with Cantera and the GRI-Mech 3.0 thermodynamic data. This is the thermodynamic ceiling against which the reactor model is checked; at 850 °C the equilibrium conversion is 0.80 and the dry hydrogen fraction 73 %.

S11. Thermal-fatigue screening of the scenarios

The life model of the main text covers creep only. Because the elastic thermal stress from the radial wall gradient is an order of magnitude above the hoop stress (Section S1), the question of whether cyclic operation adds a fatigue cost to the creep cost cannot be left open. This section screens the exposure. It is not a creep–fatigue model: no fatigue curve exists in the open literature for the alloy used here, and none is assumed. What is computed is the cyclic content of each scenario, compared with the cyclic exposure the plant already accepts from start-up and shutdown.

The hourly wall gradient is evaluated with the coupled physics model at every distinct operating point of each scenario history — 78 600 coupled solves in total — rather than through a surrogate. A Gaussian-process surrogate was trained for this quantity in a preliminary trial and discarded, and the two figures that follow come from that trial rather than from the reported pipeline: its error over the sampled operating box was 3.8 K RMSE, of the same order as the stress ranges being counted, and its noise generated 547 spurious cycles per year in a scenario that holds a single operating point all year. The thermal stress follows the relation and constants of Section S1, and the cyclic content of each annual history is extracted by rainflow counting following ASTM E1049.

Two elevations must be distinguished. The maximum of the outer-wall temperature and the maximum of the wall gradient are not at the same place, and they part company under over-firing: at a firing factor of 1.22 the outer-wall maximum migrates from $z/L = 0.43$ to the tube outlet, where the process gas has caught up with the wall and the local flux is small, so the gradient at the hot spot collapses from 57 to 15 K while the wall there is 36 K hotter. The maximum gradient anywhere on the tube instead rises monotonically with firing (63, 66, 70 K). The maximum-gradient definition is therefore the one that bounds the thermal stress and is used below; the hot-spot values, which are the ones relevant to the creep calculation, are given alongside.

One full start-up and shutdown — cold tube with no firing and no flux, hence no gradient, to the calibrated base state and back — is one cycle of the end-to-end range, 122 MPa on the maximum-gradient definition and 110 MPa at the hot spot. Reformer plants undergo of the order of 2–6 full cool-downs a year; that rate is plant practice, assumed

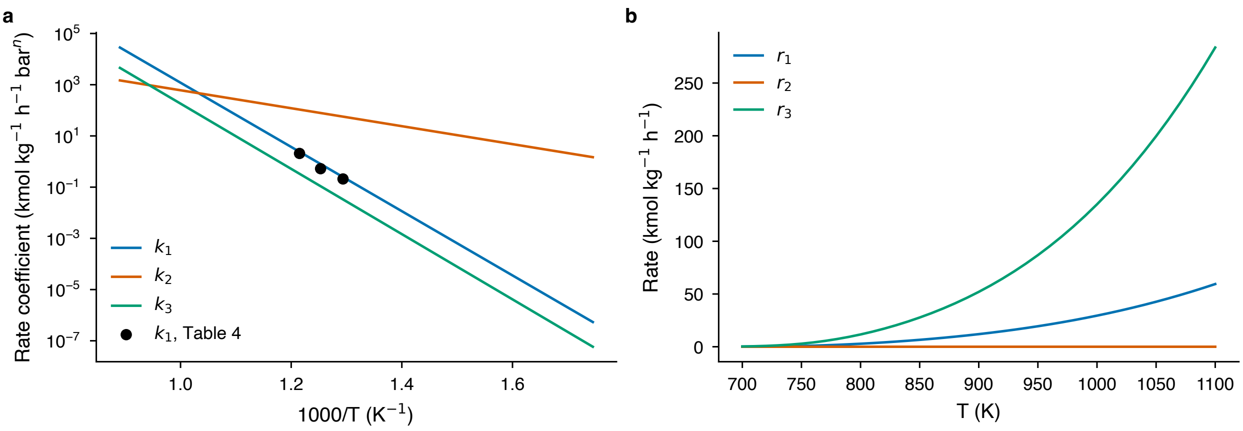

Figure S4: Xu–Froment kinetics as implemented: (a) Arrhenius plot of the rate coefficients k_1, k_2 and k_3 against $1000/T$, with the per-temperature estimates of Table 4 of [1] overlaid, and (b) the three reaction rates against temperature at 25 bar for a CH_4:H_2O:H_2 feed of 1:3:0.1.

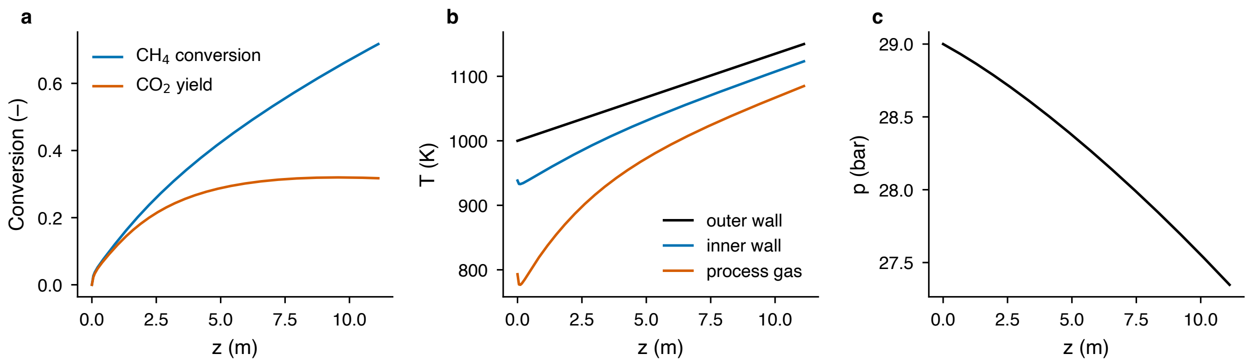

Figure S5: Stand-alone tube model with a prescribed linear outer-wall profile: (a) methane conversion and CO_2 yield, (b) process-gas, inner-wall and outer-wall temperatures and (c) pressure along the tube for the feed of Table 2 of [2]. This run precedes the verification of the main text and uses no fitted parameters.

here, not a result of the model.

The exponent decides the answer, so it is swept rather than assumed. One year of daily load-following equals the exposure of a single start-up at $m = 4.15$ on the maximum-gradient definition and $m = 4.60$ at the hot spot; it equals the assumed 2–6 start-ups per year at m between 2.9 and 3.7. Above $m \approx 4$ a year of daily cycling is worth a fraction of one start-up, and at $m = 8$ it is 0.39 % of one. No fatigue curve for this alloy is available in the open literature, so the screening is reported as a condition rather than a verdict: the cyclic exposure added by smooth load variation stays below the exposure the plant already accepts provided the fatigue exponent exceeds about 3.7, and does not below it. Establishing that exponent for centrifugally cast reformer alloys is, after the flame-length response, the second measurement this study identifies as decisive.

Three limitations are stated rather than absorbed. The elastic range is an upper bound, because in steady operation the thermal stress relaxes by creep and is conventionally excluded from the design stress. The elastic constants are single representative values, not temperature-dependent properties. And scenario S4c is invisible to this calculation by construction: its flame-impingement proxy adds 40 K directly to the metal temperature rather than changing an operating input, and a 40 K rise from additional local flux and one from redistributed radiation imply different gradients, which the scenario does not specify. Its row is a lower bound, and no flux was invented to fill it.

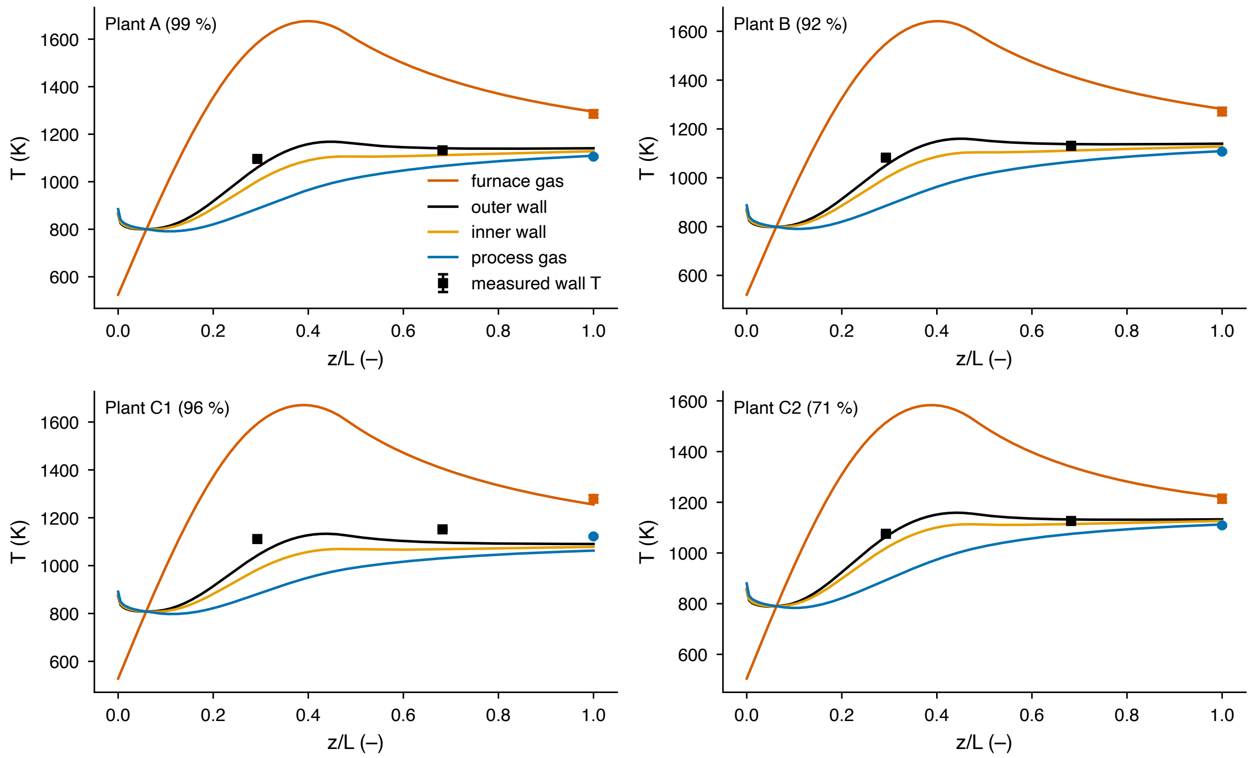

Figure S6: Coupled furnace–tube model before calibration, with the parameter values of Table 6 of [3] transferred directly: axial temperature profiles for the plant cases and the measured wall temperatures. Three of the four cases are reproduced within the measurement uncertainty without any fitting.

References

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Table S11: Thermal-fatigue screening of the annual scenarios: largest stress range and cycle count from rainflow counting of the hourly history, and the annual cyclic content expressed as a multiple of one start-up cycle under a Basquin-type exponent m. Ranges below 1 MPa are excluded from the counts; they are numerical texture of the firing bisection and are worth less than 10^{-6} of a start-up each. Values are on the maximum-gradient definition. No fatigue curve for the alloy is used: m is swept, not assumed.

Scenario	largest range (MPa)	cycles per year	$m = 3$	$m = 5$	$m = 8$
S1 steady, S6 (both), S5 constant-activity	0	0	$\leq 10^{-6}$	$\leq 10^{-9}$	$\leq 10^{-15}$
S2 daily load-following	29	365	4.98	0.28	0.0039
S3 renewable-following	49	1828	3.86	0.40	0.020
S4a mild over-firing	6	24	0.0022	4.6×10^{-6}	4.3×10^{-10}
S4b severe over-firing	13	6	0.007	10^{-4}	10^{-7}
S5 ageing (all years)	≤ 2	≤ 0.5	$\leq 10^{-6}$	$\leq 10^{-9}$	$\leq 10^{-15}$

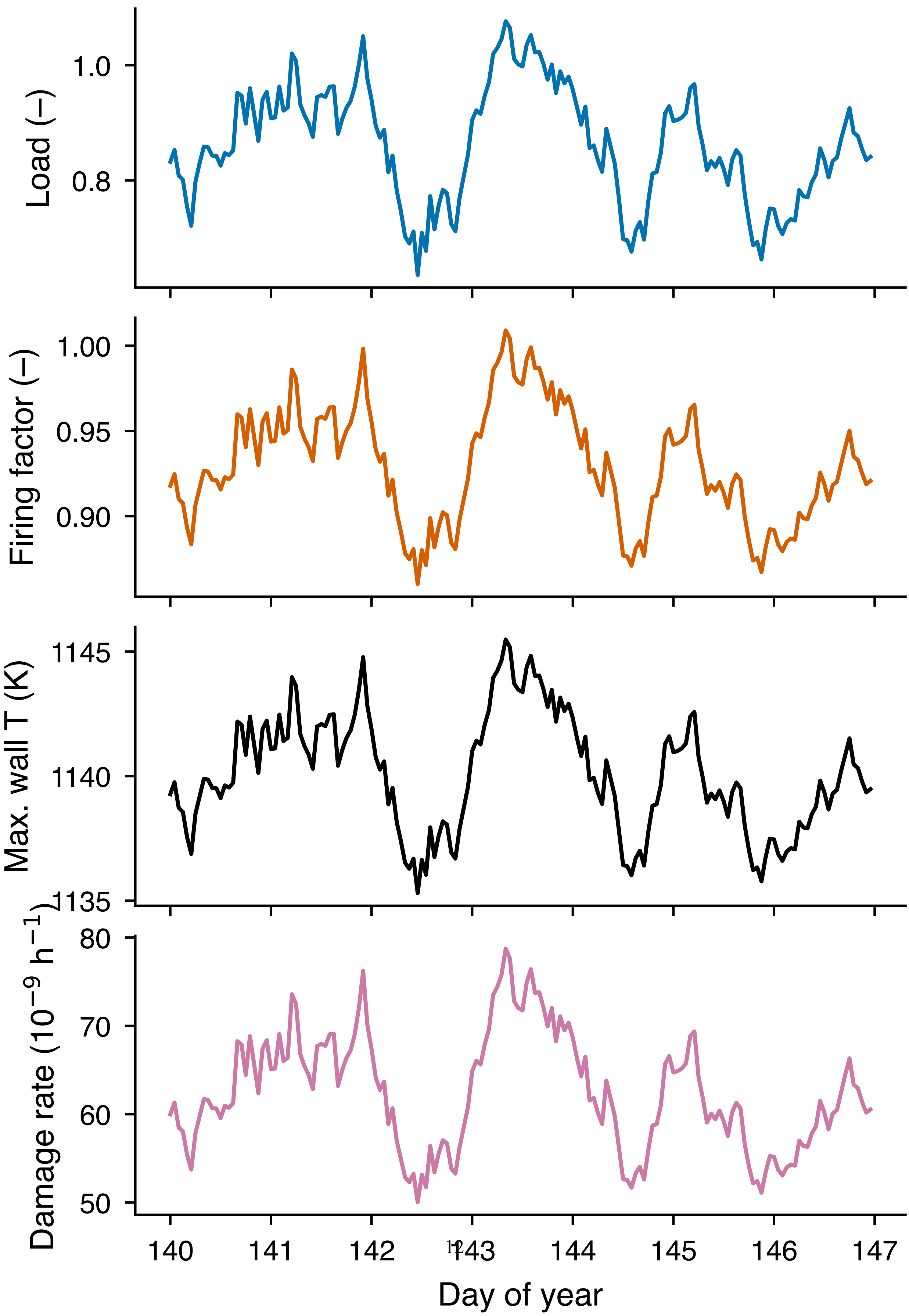

Figure S7: One representative week of the renewable-following scenario S3: load, firing factor, hot-spot temperature and hourly damage rate. The damage rate follows the hot spot through the Larson–Miller relation and is therefore strongly non-linear in the load.

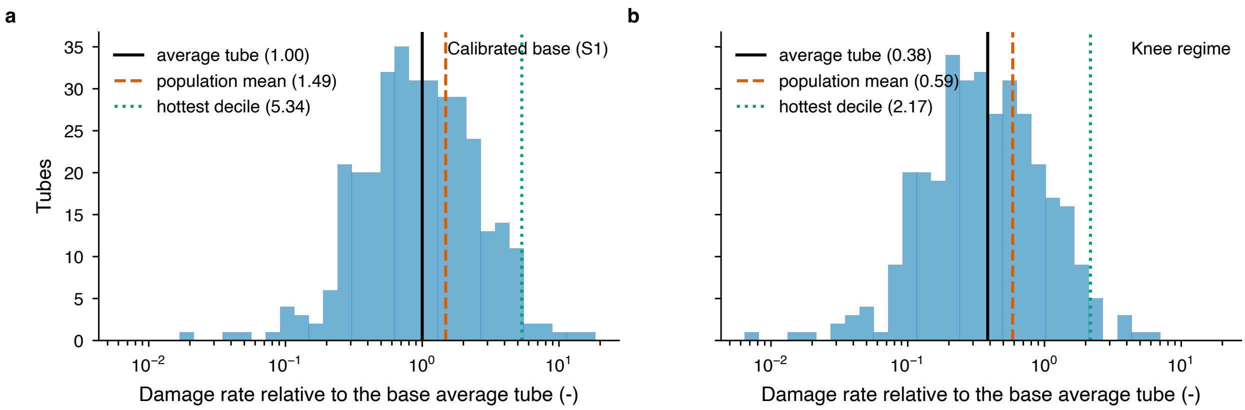

Figure S8: Distribution of the per-tube damage rate over the population of 336 tubes at $\sigma = 18.5$ K, normalized to the average tube of the base case, for (a) the calibrated base and (b) the knee regime, with the average tube, the population mean and the hottest decile marked. The whole distribution of the knee regime lies about half a decade to the left of the base.